

COS - AT1 - PSEUDO CODE WRITING TASK

1. Pseudo code: Smart transportation system

* MST: Connect all transportation infrastructure with minimum total construction cost.

SMART TRANSPORTATION(G , Source, Destination)

Input:

G = Graph of locations and roads.

Each edge has construction cost and travel distance

STEP 1.

$MST \leftarrow KRUSKAL(G)$

Step 2:

Use all roads in MST

Step 3:

Find shortest path

Step 4: Display

End

$KRUSKAL(G)$

Sort all edges

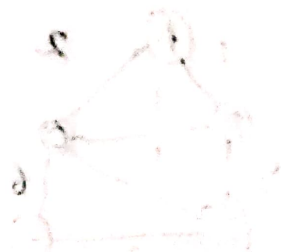
$MST \leftarrow \text{empty}$

For each edge (u, v)

If $Find(u) \neq Find(v)$:

 Add (u, v) to MST

 UNION (u, v)



If MST contains $V-1$ edges:

Break

Return MST

2. Kruskal's with Mandatory Edge Constraint.

KRUSKAL - WITH - MANDATORY ($G, \text{Mandatory Edges}$)

Create a separate set for every vertex

MST $\leftarrow$ empty.

Step 1: Add Mandatory edges

If $\text{Find}(u) = \text{Find}(v)$:

Print "ERROR".

Return ERROR

Add (u, v) to MST

UNION

TOTAL COST

Step 2:

Process remaining edges

If $\text{Find}(u) \neq \text{Find}(v)$:

Add (u, v) to MST:

UNION (u, v)

Total $+= \text{cost}(u, v)$

Step 3:

Check whether spanning tree was found

Print "Error"

Return MST, Total Cost